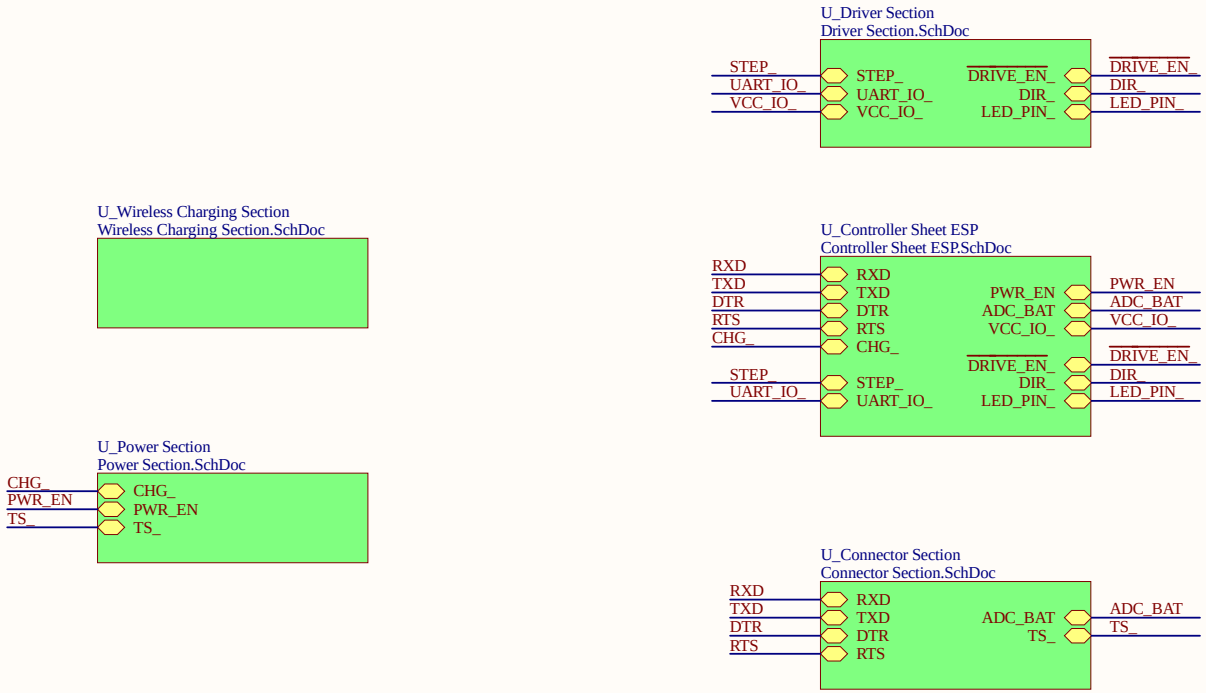
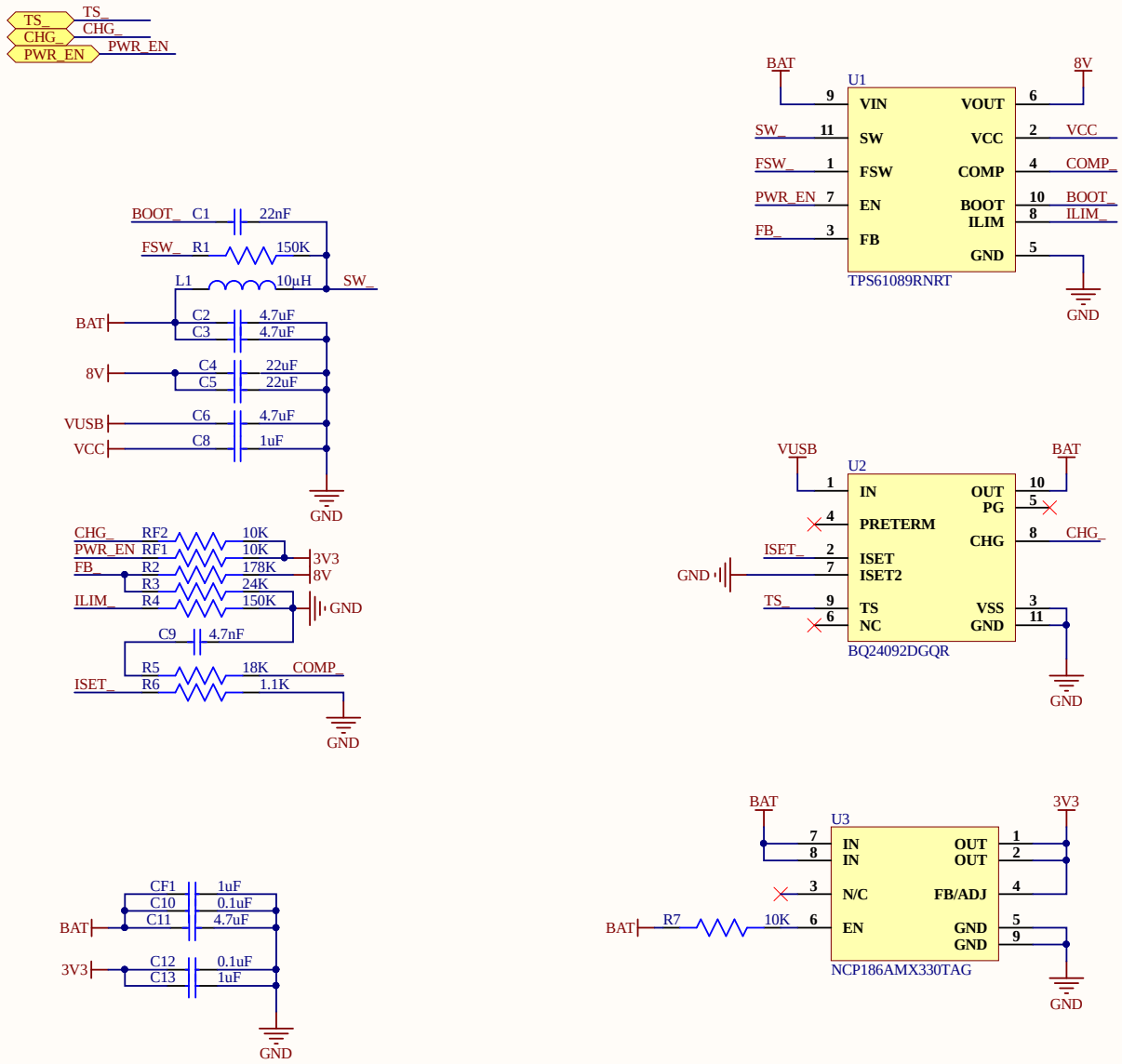
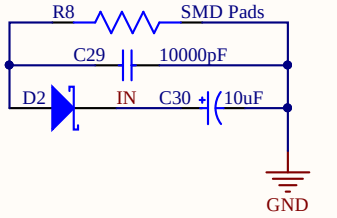
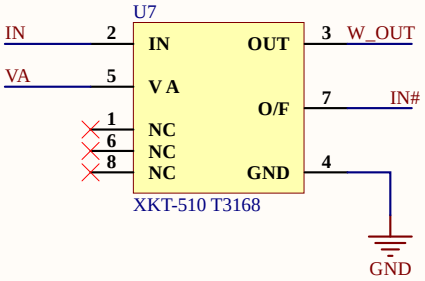

U_Driver Section
Driver Section.SchDoc
STEP_
UART_IO_
VCC_IO_
STEP_
UART_IO_
VCC_IO_
DRIVE_EN_
DIR_
LED_PIN_
DRIVE_EN_
DIR_
LED_PIN_
U_Wireless Charging Section
Wireless Charging Section.SchDoc
U_Controller Sheet ESP
Controller Sheet ESP.SchDoc
RXD
TXD
DTR
RTS
CHG_
STEP_
UART_IO_
RXD
TXD
DTR
RTS
CHG_
STEP_
UART_IO_
PWR_EN
ADC_BAT
VCC_IO_
DRIVE_EN_
DIR_
LED_PIN_
PWR_EN
ADC_BAT
VCC_IO_
DRIVE_EN_
DIR_
LED_PIN_
U_Power Section
Power Section.SchDoc
CHG_
PWR_EN
TS_
CHG_
PWR_EN
TS_
U_Connector Section
Connector Section.SchDoc
RXD
TXD
DTR
RTS
RXD
TXD
DTR
RTS
ADC_BAT
TS_
ADC_BAT
TS_

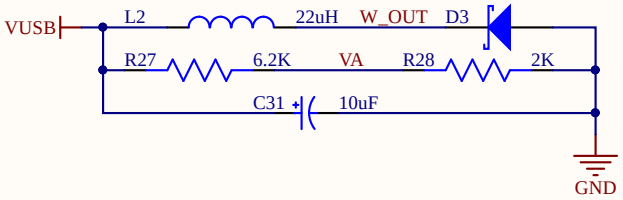

TS_
CHG_
PWR_EN
BOOT_ C1 22nF
FSW_ R1 150K
L1 10µH SW_
BAT
C2 4.7uF
C3 4.7uF
8V
C4 22uF
C5 22uF
VUSB
C6 4.7uF
VCC
C8 1uF
GND
CHG_ RF2 10K
PWR_EN RF1 10K
FB_ R2 178K
R3 24K
ILIM_ R4 150K
3V3
8V
C9 4.7nF
R5 18K COMP_
ISET_ R6 1.1K
CF1 1uF
C10 0.1uF
C11 4.7uF
C12 0.1uF
C13 1uF
U1
VIN
SW
FSW
EN
FB
VOUT
VCC
COMP
BOOT
ILIM
TPS61089RNRT
SW_
FSW_
PWR_EN
FB_
U2
IN
PRETERM
ISET
ISET2
TS
NC
OUT
PG
CHG
VSS
BQ24092DGQR
U3
IN
N/C
EN
OUT
FB/ADJ
GND
NCP186AMX330TAG
R7 10K

SMD Pads used for Wireless Charging Coil Connection.

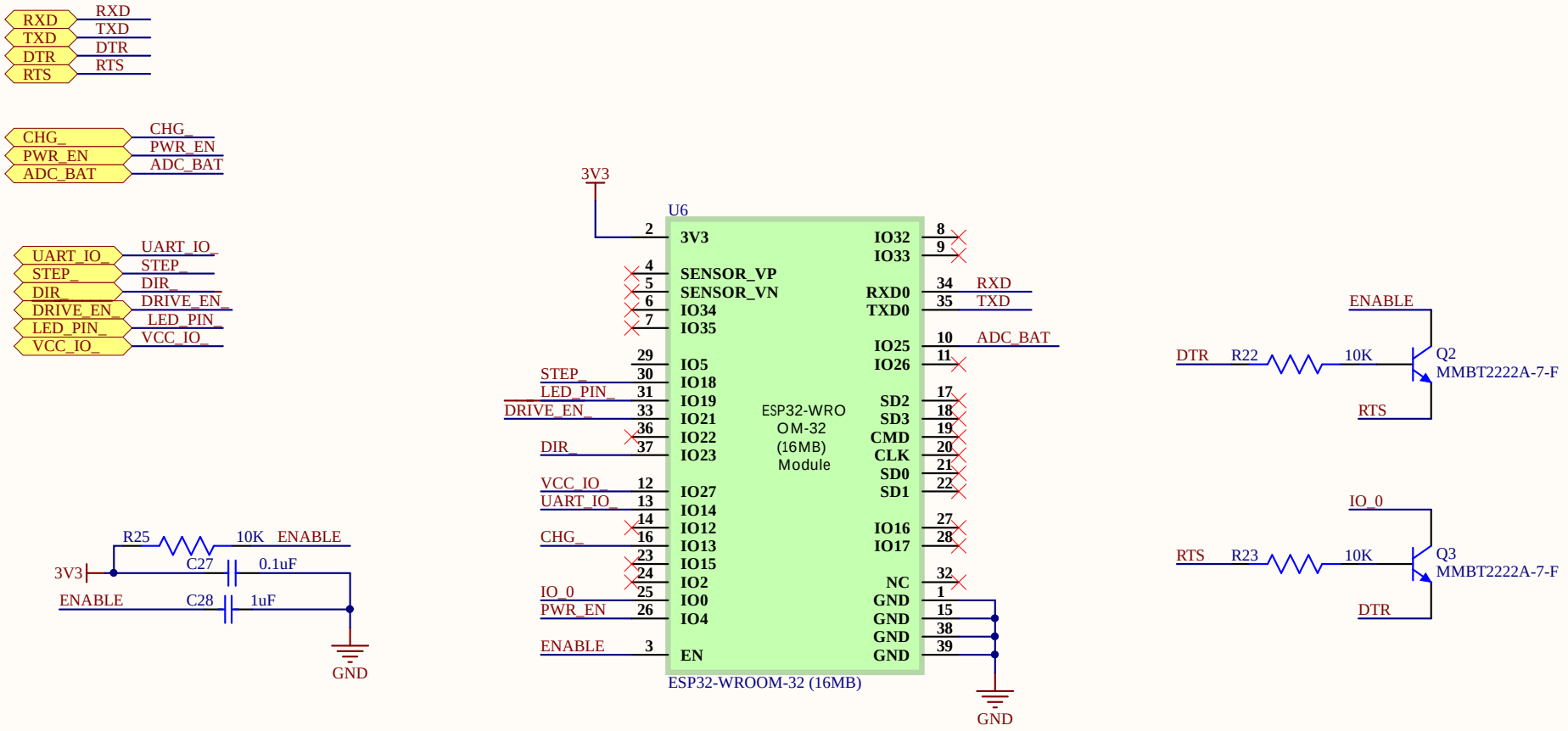




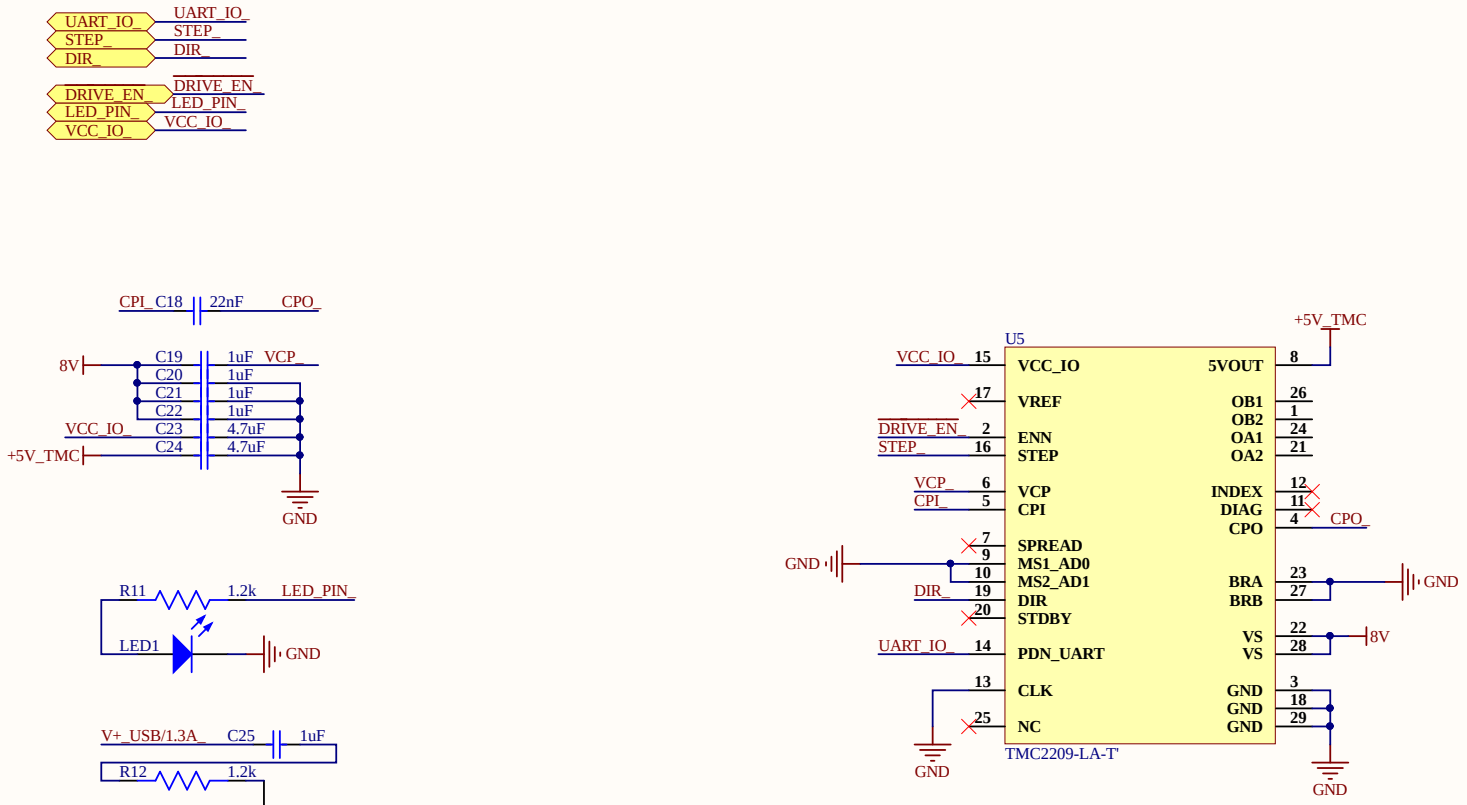


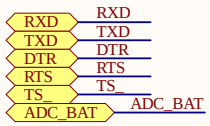


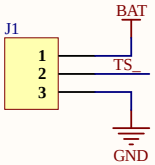
Title			
Size: A	Number:		Revision:
Date: 2/3/2021		Sheet of	
File: D:\Mars\..\Wireless Charging Section.SchDoc		Drawn By:	



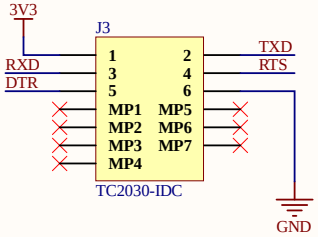
Title		
Size A	Number	Revision
Date: 2/3/2021	Sheet of	
File: D:\Mars\..\Controller Sheet ESP.SchDoc	Drawn By:	


UART_IO_
STEP_
DIR_
DRIVE_EN_
LED_PIN_
VCC_IO_
CPI_ C18 22nF CPO_
8V
C19 1uF VCP_
C20 1uF
C21 1uF
C22 1uF
VCC_IO_ C23 4.7uF
+5V_TMC C24 4.7uF
GND
R11 1.2k LED_PIN_
LED1
GND
V+_USB/1.3A_ C25 1uF
R12 1.2k
GND
Q1
SW2
U5
VCC_IO_ 15
VCC_IO
17
VREF
DRIVE_EN_ 2
ENN
STEP_ 16
STEP
VCP_ 6
VCP
CPI_ 5
CPI
7
SPREAD
9
MS1_AD0
10
MS2_AD1
DIR_ 19
DIR
20
STDBY
UART_IO_ 14
PDN_UART
13
CLK
25
NC
TMC2209-LA-T'
GND
+5V_TMC
5VOUT 8
OB1 26
OB2 1
OA1 24
OA2 21
INDEX 12
DIAG 11
CPO 4
CPO_
BRA 23
BRB 27
GND
VS 22
VS 28
8V
GND 3
GND 18
GND 29
GND


RXD
TXD
DTR
RTS
TS_
ADC_BAT
RXD
TXD
DTR
RTS
TS_
ADC_BAT


J1
1
2
3
BAT
TS_
GND


BAT
ADC_BAT


3V3
J3
RXD
DTR
1
3
5
MP1
MP2
MP3
MP4
2
4
6
MP5
MP6
MP7
TXD
RTS
GND
TC2030-IDC